pamaldi at SemEval-2026 Task 11: Neuro-Symbolic Syllogistic Reasoning via LLM-Guided Structure Extraction and Deterministic Validation

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Abstract

We describe our participation in SemEval-2026 Task 11, Subtask 1: determining the formal validity of syllogisms in English while minimizing the influence of content plausibility. Our system implements a neuro-symbolic pipeline that strictly separates the neural and symbolic components of syllogistic reasoning. An LLM extracts the formal structure of natural-language syllogisms—identifying the proposition types (A, E, I, O) and the three terms—while the syllogistic figure is computed deterministically and a symbolic validator checks whether the resulting mood–figure pair belongs to the 24 classically valid Aristotelian forms. Self-consistency voting across multiple extraction samples and targeted post-processing rules further improve robustness. On the official evaluation, our system achieves 96.34% accuracy, a Total Content Effect (TCE) of 1.02, and a combined score of 56.57. When the symbolic path decides (96.3% of instances), TCE is effectively zero by construction; the residual TCE of 1.02 is localized to the LLM fallback branch. All seven residual errors originate in the neural extraction stage, while the symbolic validator introduces no content-dependent failures. Compared to pure-LLM baselines on the same backbone, our system closes the content-effect gap without loss of accuracy.

1 Introduction

Large language models (LLMs) exhibit a persistent vulnerability to *content effects* in logical reasoning: they tend to accept logically invalid arguments when the conclusion is plausible, and reject valid arguments when the conclusion seems implausible (Dasgupta et al., 2022; Eisape et al., 2024). This entanglement of formal reasoning with world knowledge undermines the reliability of LLMs in tasks requiring strict logical inference.

SemEval-2026 Task 11 (Valentino et al., 2026) addresses this challenge by evaluating systems on syllogistic reasoning across four subtasks of increasing complexity, from English-only binary classification to multilingual settings with irrelevant premises. In this paper, we focus on **Subtask 1**: predicting the formal validity of categorical syllogisms in English, where the official ranking metric jointly rewards high accuracy and penalizes content bias through the Total Content Effect (TCE).

Our key insight is that syllogistic validity is a purely structural property: it depends entirely on the *mood* (the combination of proposition types A, E, I, O) and the *figure* (the arrangement of terms across premises), not on the truth or plausibility of the premises. This motivates a neuro-symbolic architecture that decomposes the problem into two stages: (1) a *neural extraction* stage, where an LLM parses the natural-language syllogism into its formal components, and (2) a *symbolic validation* stage, where a deterministic checker verifies whether the extracted mood–figure pair belongs to the 24 classically valid Aristotelian forms.

Our contributions are: (1) a strict neuro-symbolic decomposition in which the LLM is forbidden from producing reasoning steps and emits only four AEIO labels plus three term strings, with all inference performed by a closed-form lookup; (2) a controlled comparison, on the same backbone, against direct-prompt, chain-of-thought, self-consistent chain-of-thought, and an LLM-generates-Prolog variant, showing that reducing the LLM's reasoning role improves both accuracy and TCE; (3) a decomposition of the residual content effect showing it is localized almost entirely in the LLM fallback branch, not in the symbolic validator; and (4) a multi-model study of the extractor demonstrating that the design claim—that the symbolic stage is the invariant content-immune component—holds across backbones.

Our submitted system achieves a combined score of 56.57 with 96.34% accuracy and a TCE of only 1.02. Code, prompts, and evaluation scripts are publicly available.[1,2]

2 Background

2.1 Task Description

SemEval-2026 Task 11 (Valentino et al., 2026) investigates whether computational systems can perform formal logical reasoning independently of content plausibility. Subtask 1 requires predicting the binary validity label of syllogisms presented in English. Each syllogism consists of two premises and a conclusion, and is annotated with both a `validity` label (the target) and a `plausibility` label (reflecting real-world commonsense). The training set contains 200 instances and the test set 191 instances, each balanced across four conditions: valid–plausible (VP), valid–implausible (VI), invalid–plausible (IP), and invalid–implausible (II).

2.2 Evaluation Metrics

Systems are evaluated on three metrics: (1) **Overall Accuracy** (ACC); (2) **Total Content Effect** (TCE), a composite score measuring the average accuracy difference due to plausibility across logical conditions—a lower TCE indicates greater robustness to content bias; and (3) the **Primary Ranking Metric**, computed as

$$\text{Score} = \frac{\text{ACC}}{1 + \ln(1 + \text{TCE})}. \qquad (1)$$

This metric rewards high accuracy while smoothly penalizing content bias.

2.3 Categorical Syllogisms

A categorical syllogism consists of exactly three terms—the *major term* (P), *minor term* (S), and *middle term* (M)—distributed across two premises and a conclusion. Each proposition takes one of four forms: **A** ("All S are P"), **E** ("No S are P"), **I** ("Some S are P"), or **O** ("Some S are not P"). The *mood* is the ordered triple of proposition types (e.g. AAA, EIO); the *figure* (1–4) is determined by the positions of the middle term across the two premises (Copi et al., 2014). Under the traditional Aristotelian interpretation with *existential import*, exactly 24 mood–figure pairs are valid, distributed across the four figures.

[1] `https://github.com/pamaldi/pamaldi_semeval_task11`

[2] Language of data: English.

2.4 Related Work

Content effects in LLM reasoning have been extensively documented (Dasgupta et al., 2022; Eisape et al., 2024; Bertolazzi et al., 2024; Ozeki et al., 2024; Seals and Shalin, 2024), with mechanistic analyses of syllogistic inference circuits (Kim et al., 2025). Mitigation approaches include fine-grained activation steering (Valentino et al., 2025; Maraia et al., 2026), quasi-symbolic chain-of-thought (Ranaldi et al., 2025), faithful-CoT methods (Lyu et al., 2023; Xu et al., 2024), and LLM–symbolic theorem-prover combinations (Quan et al., 2024).

Our approach differs in a stricter sense. Whereas faithful-CoT and symbolic-CoT methods (Lyu et al., 2023; Xu et al., 2024; Ranaldi et al., 2025) and theorem-prover hybrids (Quan et al., 2024) still rely on LLM-generated reasoning steps that are validated or executed downstream, our pipeline forbids the LLM from producing any reasoning: it emits only AEIO type labels and term strings. The syllogistic figure is computed deterministically from term positions, and all inference is a closed-form lookup against the 24 valid Aristotelian forms. As we show in Section 3.1, this stricter separation yields both higher accuracy and lower TCE than an LLM-generates-Prolog variant on the same data.

3 System Overview

3.1 Design Evolution

The final system emerged from an iterative exploration of four architectures, each addressing limitations uncovered in the previous one. We re-executed all approaches on the official test set (191 instances) after gold labels were released; results are summarized in Table 1.

Prolog Code Generation with Reflexion. Our first approach prompted the LLM to generate executable SWI-Prolog programs encoding each syllogism's logical structure, with a Reflexion self-correction loop (Shinn et al., 2023) providing up to three retry attempts on execution failure. While conceptually elegant, this approach achieved only 90.58% accuracy with a TCE of 8.49 (combined score 27.87). Only 110 of 191 test instances succeeded on the first attempt, and the LLM could encode content bias directly into the generated rules.

Neuro-Symbolic with Full LLM Extraction. Observing that *validity checking does not require code generation*, we redesigned the pipeline as

Approach	Acc.	TCE	Score
Prolog + Reflexion	90.58	8.49	27.87
Incr. Meta-Learning	95.29	6.25	31.97
Full LLM Extraction	97.91	2.13	45.74
Simpl. Det. Figure	97.38	2.08	45.81
Final (submitted)	**96.34**	**1.02**	**56.57**

Table 1: Test set results (191 instances) across explored approaches. The final submitted system achieves the best combined score thanks to its lowest TCE. The first four rows are post-hoc re-evaluations on the official test set.

a two-stage system: the LLM extracts syllogistic structure (types, terms, figure) as a pure NLU task, while a deterministic module checks membership in the 24 valid Aristotelian forms. With self-consistency voting ($k = 4$), extraction reflexion, and figure verification, this variant reached 97.91% accuracy and TCE of 2.13 (combined score 45.74). Error analysis revealed two bottlenecks: incorrect figure computation by the LLM, and I/O proposition type confusion.

Simplified Extraction with Deterministic Figure. To eliminate figure errors—the most frequent category—we further reduced the LLM's role: it extracts *only* proposition types and terms, while the figure is computed *deterministically* from the middle term's position. With $k = 3$ samples and no reflexion, this reached 97.38% accuracy and a lower TCE of 2.08 (combined score 45.81).

Incremental Meta-Learning. We also experimented with automatic prompt refinement: training errors were analyzed every 50 samples and used to generate an enhanced extraction prompt with few-shot examples and guidelines (the prompt grew from 15,878 to 24,585 characters across four checkpoints). While this helped on training data, it reached only 95.29% accuracy on the test set with a TCE of 6.25 (combined score 31.97), suggesting the meta-learned prompt overfit to training error patterns.

Convergence. The submitted system combines the most effective elements: simplified NLU extraction, deterministic figure computation, self-consistency voting, targeted post-processing rules, and an LLM fallback. The central finding across all approaches is that errors consistently originate in neural extraction—never in deterministic validation—and that more deterministic components yield lower content effects.

Figure	Valid Moods
1	AAA, EAE, AII, EIO, AAI, EAO
2	EAE, AEE, EIO, AOO, EAO, AEO
3	AAI, IAI, AII, EAO, OAO, EIO
4	AAI, AEE, IAI, EAO, EIO, AEO

Table 2: The 24 valid syllogistic forms under Aristotelian logic with existential import, by figure.

3.2 Final Pipeline

The submitted system follows a two-stage neuro-symbolic pipeline: (1) the LLM extracts the syllogism's proposition types and terms as three independent samples, aggregated by majority vote; (2) post-processing rules correct common errors (I/O confusion, double negation, compound terms); (3) the syllogistic figure is computed deterministically from the middle term's position; (4) a symbolic validator checks membership in the 24 valid Aristotelian forms; (5) when structured extraction fails, a direct LLM fallback provides a validity judgment.

3.3 Neural Extraction Stage

Given a syllogism in free text, the LLM extracts the **three terms** (S, P, M) and the **proposition types** (A, E, I, O) for each of the two premises and the conclusion. The prompt includes detailed I/O distinction examples and a worked example for term identification (full template in Appendix A).

Self-Consistency Voting. We sample $k = 3$ independent extractions at temperatures 0.1, 0.3, and 0.5, selecting the most frequent mood–term combination. When all three disagree, the first extraction is used.

3.4 Symbolic Validation Stage

The deterministic checker verifies whether the extracted mood–figure pair belongs to the 24 valid Aristotelian forms (Table 2). This component contains no learned parameters and makes no reference to semantic content, ensuring complete immunity to content effects by construction.

3.5 Post-Processing and Fallback

Between extraction and validation, rule-based post-processing corrects systematic errors: **I/O disambiguation** detects negation markers to fix I↔O misclassifications; **double-negation handling** recognizes patterns like "no X are not Y" as type A; **compound-term normalization** ensures consistent matching of complex noun phrases. When structured extraction fails—i.e. none of the $k = 3$ samples yields a well-formed mood–figure pair (3.7% of test instances, 7/191)—the system falls back to a direct LLM validity judgment. As we show in Section 5.2, the fallback branch is also where the system's residual content effect is concentrated.

4 Experimental Setup

Data and Model. Subtask 1 provides 200 training and 191 test English syllogisms, balanced across four validity–plausibility conditions. We use the training set for prompt development and error analysis only; no parameters are fine-tuned. The extraction and fallback components use Qwen3-32B[3] with few-shot prompting. For self-consistency, we sample $k = 3$ extractions at temperatures 0.1, 0.3, and 0.5. The pipeline is implemented in Python 3.11 using `boto3` for LLM inference; all other components are pure Python.

Baselines. To isolate the contribution of the neuro-symbolic decomposition, we compare against three pure-LLM baselines on the same backbone (Qwen3-32B) and the same test set: **B1: Direct prompt** ("Is the following syllogism valid? Answer VALID or INVALID", temperature 0); **B2: Chain-of-thought** (same prompt plus "Think step by step, then answer on the last line", temperature 0.3, single sample); **B3: CoT + self-consistency** ($k = 4$ CoT samples at temperatures $[0.0, 0.3, 0.5, 0.7]$, majority vote). To probe cross-model generalisation of the extractor, we also run the full pipeline with the symbolic validator held fixed and two alternative backbones (Appendix D details version IDs).

Development. The system was developed through iterative error analysis on the training set. Key iterations included adding I/O disambiguation rules, restoring the full 24 Aristotelian forms (with existential import) after confirming that the task adopts the traditional interpretation, and fixing self-consistency voting and fallback parsing bugs.

5 Results

Our system achieves 96.34% accuracy (184/191), a TCE of 1.02, and a combined score of 56.57 on the official evaluation. Table 3 reports accuracy across the four validity–plausibility subgroups: performance is near-identical across all four conditions,

[3] Accessed via Amazon Bedrock.

	Plausible	**Implausible**
Valid	95.83% (46/48)	95.83% (46/48)
Invalid	97.87% (46/47)	95.83% (46/48)

Table 3: Accuracy across validity–plausibility subgroups (test set). Near-uniform performance yields TCE = 1.02; overall accuracy 96.34%, combined score 56.57.

System (Qwen3-32B)	**Acc.**	**TCE**	**Score**
B1: Direct prompt	TBD_B1_A	TBD_B1_T	TBD_B1_S
B2: Chain-of-thought	TBD_B2_A	TBD_B2_T	TBD_B2_S
B3: CoT + SC ($k = 4$)	TBD_B3_A	TBD_B3_T	TBD_B3_S
Prolog + Reflexion	90.58	8.49	27.87
Ours (final)	**96.34**	**1.02**	**56.57**

Table 4: Controlled comparison on the official test set (191 instances), same Qwen3-32B backbone. B1–B3 are pure-LLM baselines; the Prolog row reuses the LLM for code generation but still performs symbolic execution.

confirming that the symbolic validation stage successfully decouples validity judgments from content plausibility.

The errors comprise 4 false negatives and 3 false positives, spread across subgroups as (VP=2, VI=2, IP=1, II=2).

5.1 Comparison with Pure-LLM Baselines

Table 4 compares our system against the three pure-LLM baselines on the same 191-instance test set, using the same Qwen3-32B backbone. Direct prompting and chain-of-thought achieve competitive accuracy but exhibit substantially higher TCE than our system, translating into markedly lower combined scores. Adding self-consistency to CoT does not close the content-effect gap, confirming that sampling alone is insufficient: the gains come from the neuro-symbolic decomposition itself.

5.2 Fallback Branch Analysis

Of the 191 test instances, 184 are resolved via the symbolic path and 7 via the LLM fallback. Table 5 decomposes the system's accuracy and TCE by branch.

The symbolic path exhibits TCE effectively zero by construction, confirming content-invariance where the validator decides. The system-level TCE of 1.02 is therefore almost entirely produced by the fallback branch; reducing fallback frequency by improving extraction is the most direct route to lower system-level content effect.

Path	N	Acc.	TCE	Score
Symbolic only	TBD_N_S	TBD_A_S	TBD_T_S	TBD_C_S
Fallback only	TBD_N_F	TBD_A_F	TBD_T_F	TBD_C_F
System (union)	191	96.34	1.02	56.57

Table 5: Per-branch decomposition on the test set. The symbolic path is content-invariant by construction; the system-level TCE is attributable to the fallback branch, which handles 3.7% of instances.

Extractor	Acc.	TCE	Score
Qwen3-32B (submitted)	96.34	1.02	56.57
TBD_MODEL_A	TBD_A_A	TBD_A_T	TBD_A_S
TBD_MODEL_B	TBD_B_A	TBD_B_T	TBD_B_S

Table 6: Swapping the extractor LLM while holding the symbolic validator fixed. Accuracy tracks extractor quality; TCE remains low across backbones, supporting the claim that the symbolic stage is the invariant content-immune component.

5.3 Multi-Model Extraction

To test whether the design claim—that the symbolic stage is the invariant, content-immune component—holds across backbones, we re-run the pipeline with the symbolic validator held fixed while swapping the extractor LLM. Table 6 reports results for three backbones on the full 191-instance test set.

5.4 Ablation Study

Table 7 reports the impact of key components on the training set (200 instances). Restoring the full 24 Aristotelian forms (existential import) and adding self-consistency voting yield the largest individual gains; I/O post-processing provides a modest but consistent improvement.

5.5 Paraphrase Probe

To probe sensitivity of extraction to surface form, we paraphrased 30 test instances (active↔passive voice, synonym substitution for the middle term, and reordering of premises) and re-ran the full pipeline. Accuracy changes by **TBD_PP_DELTA** percentage points; all observed errors trace to the extraction stage. This confirms that the symbolic validator is invariant to surface form and that the extractor is the appropriate locus of future improvement.

6 Error Analysis

All 7 prediction errors on the test set are attributable to the neural extraction stage; the symbolic validator introduces no errors on correctly extracted structures. The errors comprise 4 false negatives (valid syllogisms predicted as invalid) and 3 false positives (invalid syllogisms mapped to valid forms due to proposition-type misidentification).

Configuration	Acc.	TCE
Full system ($k = 3$)	97.0%	2.78
– self-consistency ($k = 1$)	94.0%	4.95
– I/O post-processing	96.0%	$\sim$4.0
– 24 → 15 valid forms	93.0%	$\sim$5.5
Baseline (initial prompt)	70.0%	>10

Table 7: Ablation study on the training set (200 instances). "–" indicates removal of a component. "– 24 → 15 valid forms" replaces the Aristotelian set with its existential-import-free reduction.

Responding to reviewer feedback, we grouped the 7 failures by their dominant linguistic trigger (Table 9):

- **AEIO type confusion (4/7)**: I↔O or A↔E mislabelling, typically triggered by negation polarity or quantifier scope.
- **Middle-term / figure identification (2/7)**: the extractor selects the wrong surface phrase as the middle term, propagating into a wrong figure.
- **Complex noun phrase (1/7)**: compound nominals that the extractor splits or merges incorrectly.

All three categories stress the neural extraction stage; the symbolic validator is insensitive to surface form.

Crucially, the 7 errors are distributed nearly uniformly across plausibility conditions (VP=2, VI=2, IP=1, II=2), confirming that extraction errors are driven by linguistic complexity rather than content plausibility. This is what yields the near-zero TCE of 1.02.

Training-set annotation issues. During development we identified 5 annotation inconsistencies in the training set where universally valid Aristotelian forms (e.g. *Celarent* EAE-1, *Baroco* AOO-2) were labelled as invalid. These cases were communicated to the task organizers and do not appear in the test set. This observation is a positive side-effect of the interpretability of the neuro-symbolic paradigm: because every prediction is explainable by a mood–figure pair, labelling anomalies are easy to detect.

7 Discussion

The strict separation of neural extraction from symbolic validation offers three advantages: (1) the symbolic component is provably immune to content effects; (2) errors are localizable to either extraction or validation; (3) every prediction is interpretable via the extracted mood–figure pair.

Existential import. A critical design decision was adopting the Aristotelian interpretation with existential import (24 valid forms) rather than modern-logic (15 forms). Ablations with 15 forms showed substantially lower accuracy ($\sim$93%), confirming that the task adopts the traditional convention.

Content effect as a structural guarantee. In principle, a perfect symbolic validator achieves TCE $= 0$. Our TCE of 1.02 is close to this minimum; the Section 5.2 decomposition shows that the residual is concentrated in the fallback branch, not in the symbolic path. Further lowering TCE therefore does not require changes to the validator but rather tighter extraction or a smaller fallback footprint.

Generalisation beyond categorical syllogisms. The symbolic validator is tied to the 24 Aristotelian categorical forms. The decomposition itself, however, is form-agnostic: extending the system to propositional or first-order fragments requires swapping the validator (e.g. a SAT/SMT backend) while preserving the extract-then-validate separation. The neural stage's role—mapping natural language to a formal structure—remains the same.

Cross-model robustness. The multi-model results (Section 5.3) indicate that accuracy tracks extractor quality while TCE remains low across backbones, which is consistent with the architectural claim that content-invariance is contributed by the symbolic stage rather than any particular LLM.

8 Conclusion

We presented a neuro-symbolic system for SemEval-2026 Task 11 that achieves 96.34% accuracy, a TCE of 1.02, and a combined score of 56.57, demonstrating near-complete immunity to content effects. Controlled comparisons on a shared backbone show that reducing the LLM's reasoning role—rather than adding sampling or reflexion on top of an LLM reasoner—is what closes the content-effect gap. A branch-level decomposition confirms that the symbolic path is content-invariant by construction and that the residual TCE is localized to the LLM fallback. A multi-model extraction study supports the claim that content-invariance is contributed by the symbolic stage rather than any particular backbone. Future work includes extending the validator to propositional and first-order fragments, reducing the fallback footprint, and adapting the pipeline to the multilingual and irrelevant-premise subtasks.

Limitations

Our system is evaluated only on Subtask 1 (English binary classification over categorical syllogisms). It does not address the multilingual setting of Subtask 3, the irrelevant-premise conditions of Subtasks 2 and 4, or non-categorical logical fragments. The symbolic-path accuracy depends on the extractor LLM's ability to parse noun phrases and quantifier scope; paraphrases and double negations remain the dominant extraction failure mode (Section 6). The LLM fallback, invoked on 3.7% of test instances, is the only component that can reintroduce content sensitivity; reducing fallback frequency by improving extraction is the most direct way to lower system-level TCE. Finally, our evaluation is limited to the 191 official test instances; a paraphrase probe on a 30-instance subsample (Section 5.5) provides only a preliminary indication of robustness to surface-form variation.

Ethical Considerations

This work uses the SemEval-2026 Task 11 dataset, which consists of synthetic categorical syllogisms in English with no personally identifying information. The system performs binary formal-validity classification and is not designed for user-facing deployment; we do not foresee direct risks of misuse. The extraction and fallback components rely on an external LLM (Qwen3-32B via Amazon Bedrock), whose training data we do not control; any biases in that model may influence extraction on rare linguistic constructions. By design, however, the symbolic validator is insensitive to such biases, and the branch-level decomposition (Section 5.2) makes the residual content sensitivity measurable.

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A Extraction Prompt Template

The extraction prompt instructs the LLM to:

1. Identify the two premises and the conclusion.
2. For each proposition, determine the type (A, E, I, or O), paying special attention to the distinction between I (“Some S are P”) and O (“Some S are *not* P”).
3. Identify the three terms (S, P, M) based on their distribution: S appears in the minor premise and the conclusion; P appears in the major premise and the conclusion; M appears in both premises but not in the conclusion.

It.	Change	Acc.	Score
0	Baseline prompt	70%	20
1	Structured extraction	90%	33
2	Self-consistency voting	94%	33.8
3	I/O post-processing	96%	40
4	24 Aristotelian forms	97%	41.7
5	Compound-term handling	97.5%	45
6	Double-negation handling	98%	48

Table 8: Development progression on the training set (200 instances). Values are approximate.

#	Type	Sub.	Gold / Pred.	Trigger
1	FN	VI	EIO-4 / TBD_P1	middle-term amb.
2	FN	VP	AAA-1 / extr. fail	compound NP
3	FN	VI	EAO-1 / TBD_P3	E→A confusion
4	FP	II	TBD_G4 / TBD_P4	conclusion polarity
5	FN	VP	AEO-4 / TBD_P5	figure misid.
6	FP	II	TBD_G6 / AOO-2	double negation
7	FP	IP	TBD_G7 / TBD_P7	A/E confusion

Table 9: All 7 test set errors with dominant linguistic trigger. FN=false negative (valid→invalid), FP=false positive (invalid→valid). Subgroups: VP/VI/IP/II as defined in Section 2.1. Gold/Pred. pairs marked *TBD* are to be verified against extractor logs.

4. Output the proposition types for the two premises and the conclusion (e.g. "EIO"), together with the three terms in a structured format.

The syllogistic figure (1–4) is then computed deterministically from the position of M in the two premises:

- Figure 1: M is subject in P1, predicate in P2.
- Figure 2: M is predicate in both premises.
- Figure 3: M is subject in both premises.
- Figure 4: M is predicate in P1, subject in P2.

B Complete List of Development Iterations

Table 8 summarizes the progression of system performance through iterative development on the training set.

C Test Set Error Details

Table 9 lists all 7 errors on the official test set with their dominant linguistic trigger.

D Multi-Model Extractor Details

The multi-model study (Section 5.3) uses the following backbones, all accessed via Amazon Bedrock with the same prompt template and sampling configuration ($k = 3$, temperatures 0.1, 0.3, 0.5):

- Qwen3-32B (*the submitted system's extractor*)
- TBD_MODEL_A (model ID: `TBD_MODEL_A_ID`)
- TBD_MODEL_B (model ID: `TBD_MODEL_B_ID`)

The symbolic validator, post-processing rules, and fallback handler are identical across all three configurations.